

WEEK 2 (Day 3 & Day 4)

Sets in Python

A set is an unordered, mutable collection of unique elements. It does not allow duplicate values and is created using curly braces `{}` or the `set()` function

```
fruits = {"apple", "banana", "mango", "orange"}

print(fruits)
print(type(fruits))

{'mango', 'apple', 'orange', 'banana'}
<class 'set'>
```

Common Set Operations

1. Adding an Element :Adds a new item to the set.

```
fruits.add("kiwi")
print(fruits)

{'apple', 'mango', 'orange', 'banana', 'kiwi'}
```

2. Updating a Set:Adds multiple elements to a set.

```
fruits.update(["grapes", "pineapple"])
print(fruits)

{'pineapple', 'apple', 'mango', 'grapes', 'orange', 'banana', 'kiwi'}
```

3. Removing an Element:Removes an element from the set.

```
fruits.remove("banana")
print(fruits)

{'pineapple', 'apple', 'mango', 'grapes', 'orange', 'kiwi'}
```

4. Discarding an Element:Removes an element without giving an error if it doesn't exist.

```
fruits.discard("apple")
print(fruits)

{'pineapple', 'mango', 'grapes', 'orange', 'kiwi'}
```

5. Length of a Set:Returns the number of elements.

```
print(len(fruits))

5
```

6. Checking Membership:Checks whether an element exists.

```
print("kiwi" in fruits)

True
```

7. Traversing a Set:Visits every element one by one.

```
for fruit in fruits:
    print(fruit)

pineapple
mango
grapes
orange
kiwi
```

8. Union of Sets :Combines two sets without duplicates.

```
A = {1,2,3}
B = {3,4,5}

print(A | B)

{1, 2, 3, 4, 5}
```

9. Intersection of Sets:Returns common elements.

```
print(A & B)
```

```
{3}
```

10. Difference of Sets:Returns elements present in the first set only.

```
print(A - B)
```

```
{1, 2}
```

Functions in Python

A function is a reusable block of code that performs a specific task. It is created using the `def` keyword.

```
def function_name():  
    print("Hello")
```

```
function_name()
```

```
Hello
```

Types of Functions

1. Function With Parameters

```
def greet(name):  
    print("Hello", name)
```

```
greet("Khushi")
```

```
Hello Khushi
```

2. Function With Return Value

```
def add(a, b):  
    return a + b  
  
result = add(10, 20)  
print(result)  
  
30
```

Common Function Concepts

1. Calling a Function: Executes the function.

```
def hello():  
    print("Python")  
  
hello()  
  
Python
```

2. Default Arguments: Assigns default values to parameters.

```
def greet(name="Guest"):  
    print("Hello", name)  
  
greet()  
greet("Khushi")  
  
Hello Guest  
Hello Khushi
```

3. Keyword Arguments: Passes values using parameter names.

```
def student(name, age):  
    print(name, age)  
  
student(age=20, name="Khushi")  
  
Khushi 20
```

4. Variable-Length Arguments (*args): Accepts multiple positional arguments.

```
def numbers(*num):  
    print(num)  
  
numbers(10,20,30,40)  
  
(10, 20, 30, 40)
```

5.Lambda Function: Creates a small anonymous function.

```
square = lambda x: x*x  
  
print(square(5))  
  
25
```

6.Recursive Function:A function that calls itself.

```
def factorial(n):  
    if n == 1:  
        return 1  
    return n * factorial(n-1)  
  
print(factorial(5))  
  
120
```

